

The University of Nottingham *Malaysia Campus*

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, REASSESSMENT 2019-2020

MATHEMATICS FOR COMPUTER SCIENTISTS (COMP1017)

Time allowed ONE hour AND THIRTY minutes (Timed exam)

Open-book examination.

Answer 3/ 3 questions

Suggested time to complete the paper ~2 hours including downloading exam questions, scanning and uploading answer scripts.

This open-book examination will be marked out of 100.

You can either write your answers on paper and scan it in as one PDF file or type your answers in a word processor and print it as one PDF file.

Submit your answers containing all the work you wish to have marked as a **single PDF file, with each page in the correct orientation, to the appropriate dropbox on the module's Moodle page.**

Use the standard naming convention for your document: [Student ID]_[Module Code]. Write your student ID number at the top of each page of your answers. Do not include your name.

Although you may use any notes or resources you wish to help you complete this open-book examination, the academic misconduct policy that applies to your coursework also applies here. You must be careful to avoid plagiarism, collusion or false authorship.

Staff are not permitted to answer assessment or teaching queries during the period in which your open-book examination is live. If you spot what you think may be an error on the exam paper, note this in your submission but answer the question as written.

0 marks shall be awarded for any late submission of this work.

1. Questions on Logic, Sets, Functions, and Relations.

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(i) $(x \neq 3)$ is a proposition.(ii) Consider a relation R where $\exists a, b : (a, b) \in R \Rightarrow (b, a) \in R$. R is symmetric.(iii) Consider $y = \pm \sqrt{\frac{4-x}{a}}$. y is an onto function of x .(iv) $\{0, 2, 12, 36, 80\} \neq \{y \in \mathbb{N} \mid \exists x \in \mathbb{N} ((y = x^2 + x^3) \wedge (0 \leq x \leq 5))\}$.(v) Propositional operator $\vee$ is a function mapping propositions to propositions.(b) Let $A = \{a, b, c, d, e\}$. Let S be the set of all functions on A , and T the set of all functions on the power set 2^A .(i) Provide an example of (x, y) for $y = g(x)$ where $g \in T$. Explain your answer.

[4 marks]

(ii) If all functions in S and T are total, determine $|S|$ and $|T|$. Explain your answer briefly.

[5 marks]

(iii) Let $f_1 = \{(a, b), (b, b), (c, e), (d, d), (e, b)\}$ and $f_2 = \{(a, d), (b, e), (e, c)\}$ where $f_1, f_2 \in S$. Determine $f_1 \circ f_2$ and explain whether it is one-to-one and whether it is onto.

[5 marks]

(iv) Define a function $g \in T$ with suitable set operator(s), such that the range of g is $\{\{d\}, \emptyset\}$, where $\emptyset$ is null set.

[6 marks]

2. Questions on Number Theory, Matrix, and Mathematical Induction.

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

- (i) Let a and $b \geq 1$ be integers. There exist unique integers q and r such that $a = bq + r$.
- (ii) Let A, B , and C be matrices. If $AC = BC$, then $A = B$.
- (iii) Let $\{a_n\}, n \geq 1$ be a sequence of numbers, then $\forall k \in \mathbb{Z}, \sum_{n=5}^{10} a_n = \sum_{n=k-10}^{k-5} a_{k-n}$.
- (iv) Let g be greatest common divisor of integers m, n . Let $g = 2^{a_1} \cdot 3^{a_2} \cdot 7^{a_3} \dots$ be written in its prime factorization form. If some $a_i = 1$ and all others are 0, then g is a prime number.
- (v) $\gcd(252, 489) = \gcd(15, 10)$.

(b) Let p be a prime. For integers a, b , if $p|ab$, then $p|a$ or $p|b$. Let $P(n)$ be the predicate that if $p|a_1 a_2 \dots a_n$, then $p|a_i$ for some i , where each a_i is an integer. Using mathematical induction, answer the following questions.

- (i) State $P(1)$ and prove it is true.

[2 marks]

- (ii) State the inductive step $P(k) \rightarrow P(k+1)$ for some integer $k \geq 1$.

[2 marks]

- (iii) Prove $P(k) \rightarrow P(k+1)$ is true using (i), (ii), and the information given above.

[7 marks]

(c) Let $\{p_n\}, 1 \leq n \leq 9$ be the sequence of the first 9 prime numbers and be stored in a 3×3 matrix A . Let $C = AA^t$. Show that the sum of the diagonal entries of C is equivalent to $\sum_{n=1}^9 p_n^2$.

[4 marks]

3. Questions on Counting and Probabilities.

(a) For each statement, say whether it is **true** or **false**.

[10 marks]

- (i) Throw an unbiased coin 5 times and the probability of obtaining 2 heads is $\frac{1}{2} \times \frac{1}{2}$.
- (ii) Put 4 apples into 2 boxes. The probability of one of the boxes having exactly 2 apples is 1.
- (iii) Hariz has either coffee or tea for breakfast but never both. He has coffee 40% of the time. Therefore, Hariz having coffee for breakfast is independent of him having tea for breakfast.
- (iv) The probability of a student studying hard but failing the subject is 0.1. The probability of the student failing the subject but not studying hard is 0.6. So the probability of the student failing the subject is 0.7.
- (v) Roll a dice of 6 sides with sides numbered from 1 to 6. The probability of observing a number that is greater than 0 is 1.

(b) There are 400 students in a programming class. On the midterm test each student received one of the following grades: *A, B, C*, or *D*.

- (i) Show that at least two of the students in the class were born on the same day of a month.

[6 marks]

- (ii) There are less students born in the first half of a year than in the latter. 60% of the grade *A* students were born in the first half of a year. Under what circumstances can one say that the event of students born in the first half of a year is independent of the event of students obtaining grade *A*. Explain your answer.

[9 marks]